

FFT ASIC Design & Implementation

ADI Summer 26' Digital IC Design Internship Final Project
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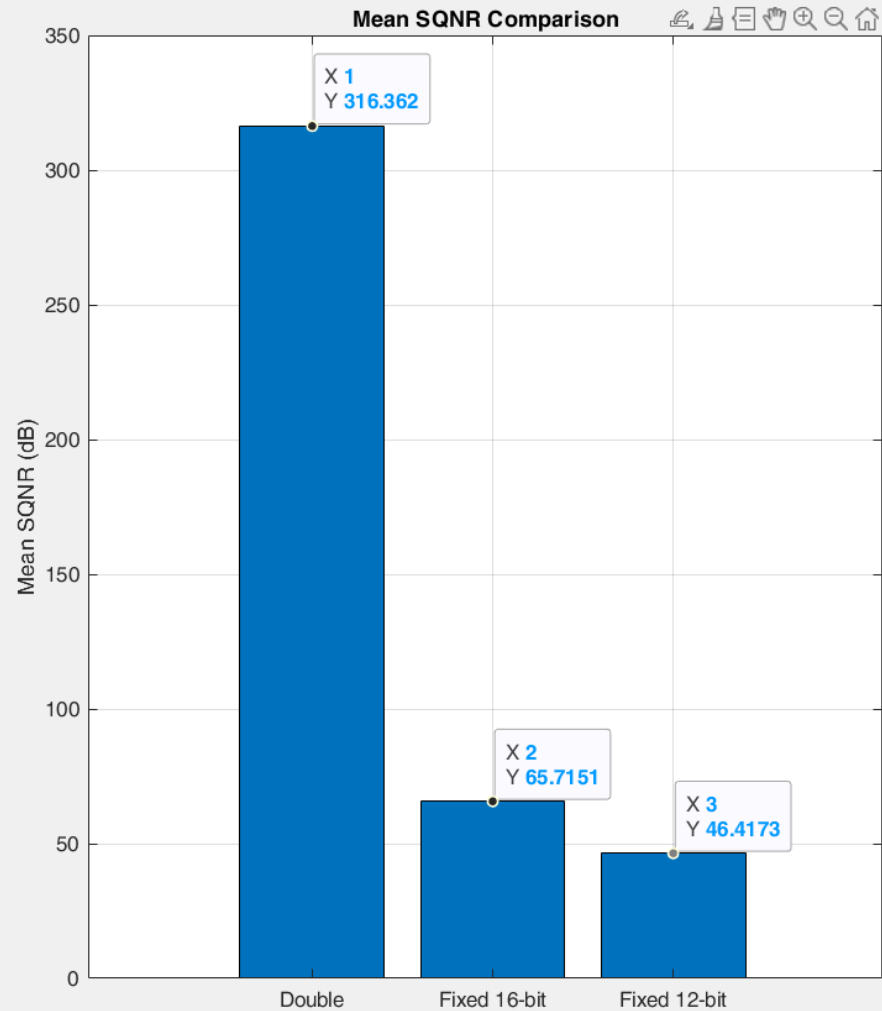
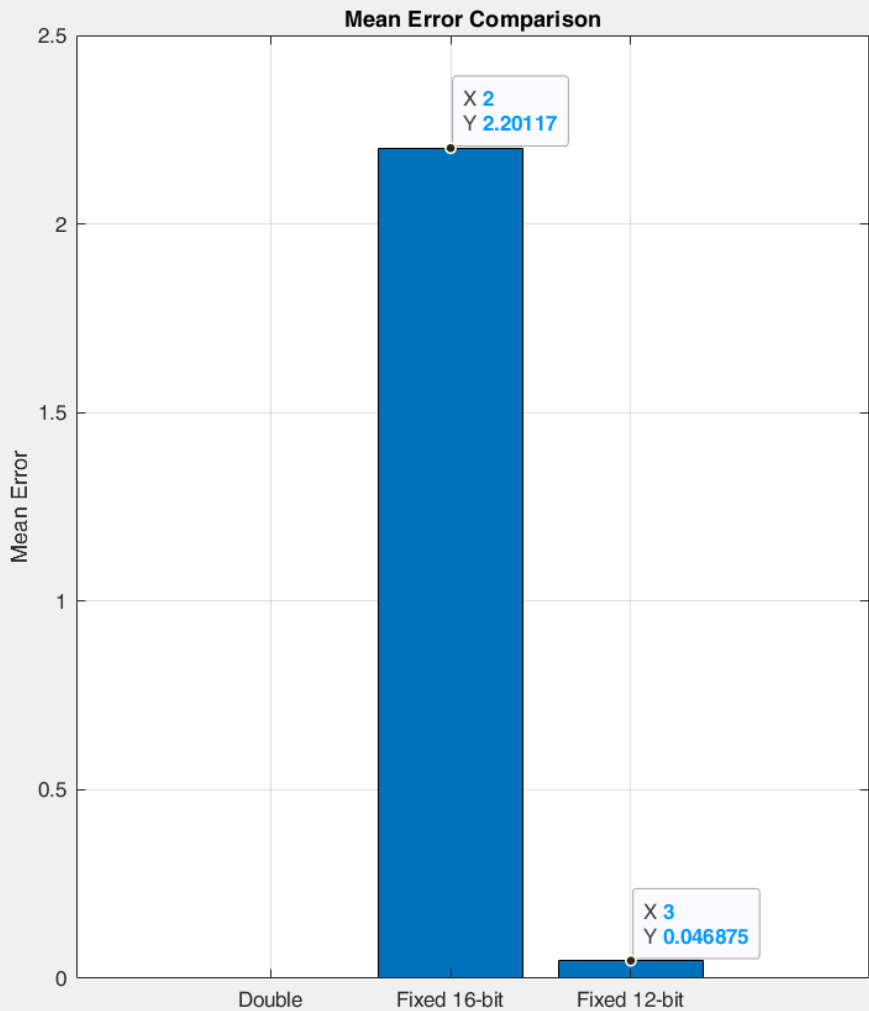
- ▶ System Modeling
- ▶ RTL Design
- ▶ RTL Verification
- ▶ ASIC Implementation
- ▶ FPGA Implementation

System Modeling

Phase 1

- ▶ Algorithms : DIT VS DIF
- ▶ Implemented 16-point Radix-2 DIF FFT Using Single-path Delay Feedback (SDF) architecture
- ▶ Initially , we model 16-point DIF algorithm using Floating Point representation
- ▶ Developed fixed-point model with optimized bit-width allocation for each stage
- ▶ Compared fixed-point implementation with floating-point implementation
- ▶ Metrics of Comparison : SQNR & Error

Results Comparison



```
===== Double Precision =====  
Mean Error : 1.870823e-15  
Max Error  : 5.024296e-15  
Mean SQNR  : 316.36 dB  
===== Fixed-Point (16-bit) =====  
Mean Error : 2.201172e+00  
Max Error  : 1.600195e+01  
Mean SQNR  : 65.72 dB  
===== Fixed-Point (12-bit) =====  
Mean Error : 4.687500e-02  
Max Error  : 7.812500e-02  
Mean SQNR  : 46.42 dB
```

- We choose 12-bit Fixed-point implementation with different integer and fraction bits for each stage

- ▶ **Number of FFT Points:** 16
- ▶ **Architecture:** DIF R2 SDF
- ▶ **Input order:** Natural order
- ▶ **Output order:** Bit-reversed order
- ▶ **Implementation:** 4 stages
- ▶ **Twiddle Factors:** Pre-computed and quantized to fixed-point representation (12-bit)

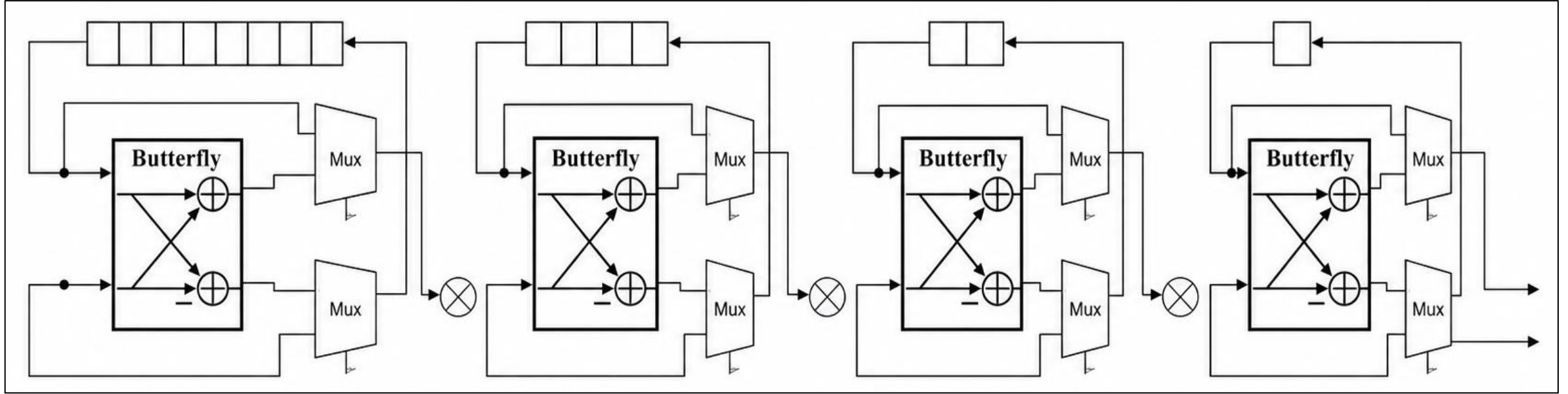
► **Word length:** 12-bit

► **Bit-Width Allocation :**

- Input signals: 4 integer + 8 fraction bits (Q4.8)
- Twiddle Factors: 2 integer + 10 fraction bits (Q2.10)
- Stage 1: 5 integer + 7 fraction bits (Q5.7)
- Stage 2: 5 integer + 7 fraction bits (Q5.7)
- Stage 3: 5 integer + 7 fraction bits (Q5.7)
- Stage 4: 6 integer + 6 fraction bits (Q6.6)
- Final Output: 6 integer + 6 fraction bits (Q6.6)

► **Rounding Method :** Flooring

SDF Architecture Implementation



► Features:

- Sequential processing
- Optimized memory usage with feedback paths.
- Butterfly operations followed by twiddle factor multiplication.
- Delay elements halve in size with each stage ($8 \rightarrow 4 \rightarrow 2 \rightarrow 1$).

RTL Design

Phase 2

► **Modular Design:**

- Four cascaded pipeline stages
- Each stage implements one level of the FFT butterfly operations
- Parameterized bit-widths for flexible precision control

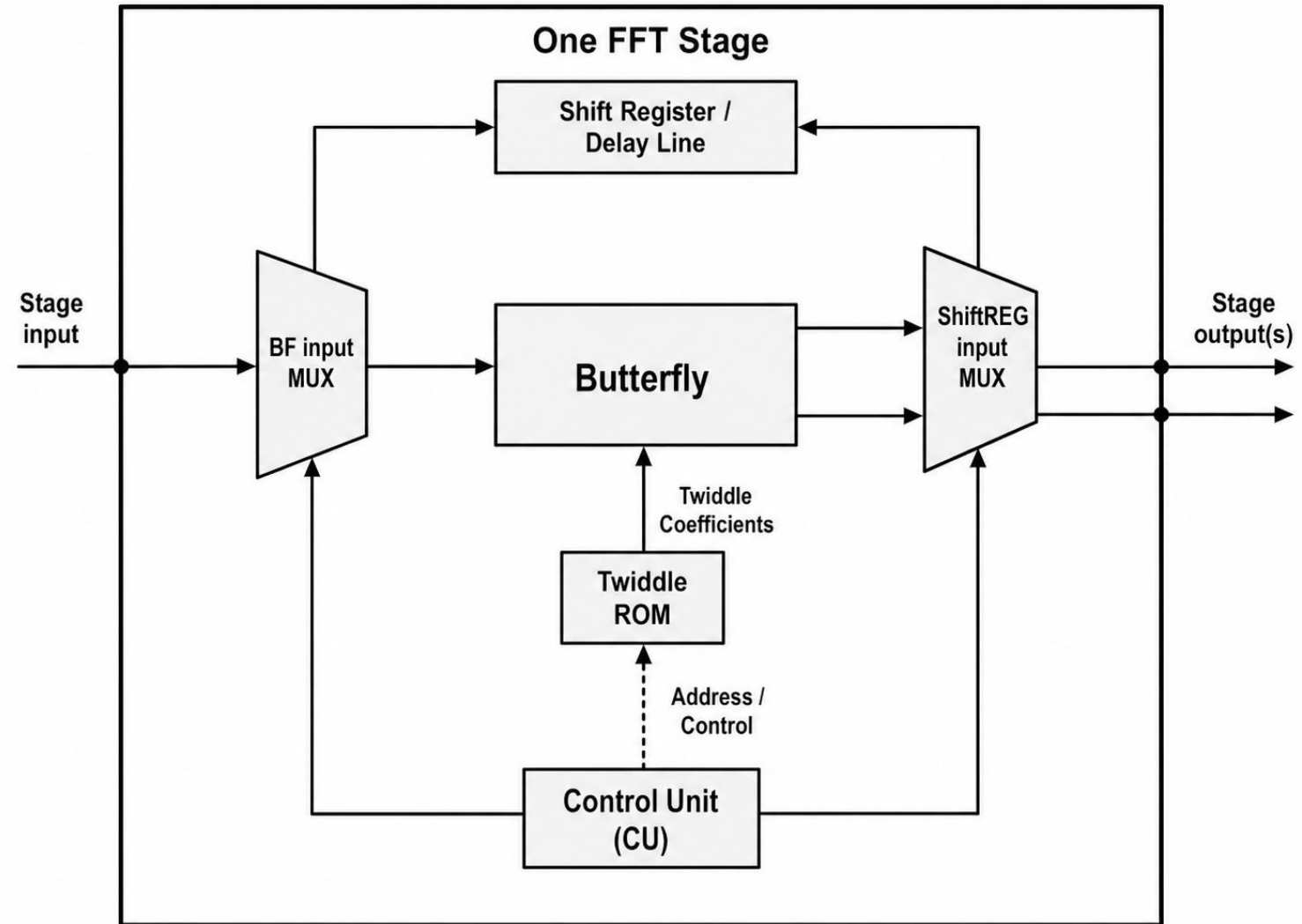
► **Control Signals:**

- First stage has invalid input signal to indicate that there is an input data
- Each stage sends valid signal to the next stage to start its operation
- Final stage has outvalid signal asserted when the output is ready

Stage Architecture

► Common stage architecture:

- Shift Register with stage-specific length
- Butterfly unit for addition /subtraction operations and instantiate complex multiplier
- ROM contain Twiddle Factor
- MUXs for input of the butterfly and the input of the shift register
- CU for MUXs selections and send addresses for the ROM



► Latency:

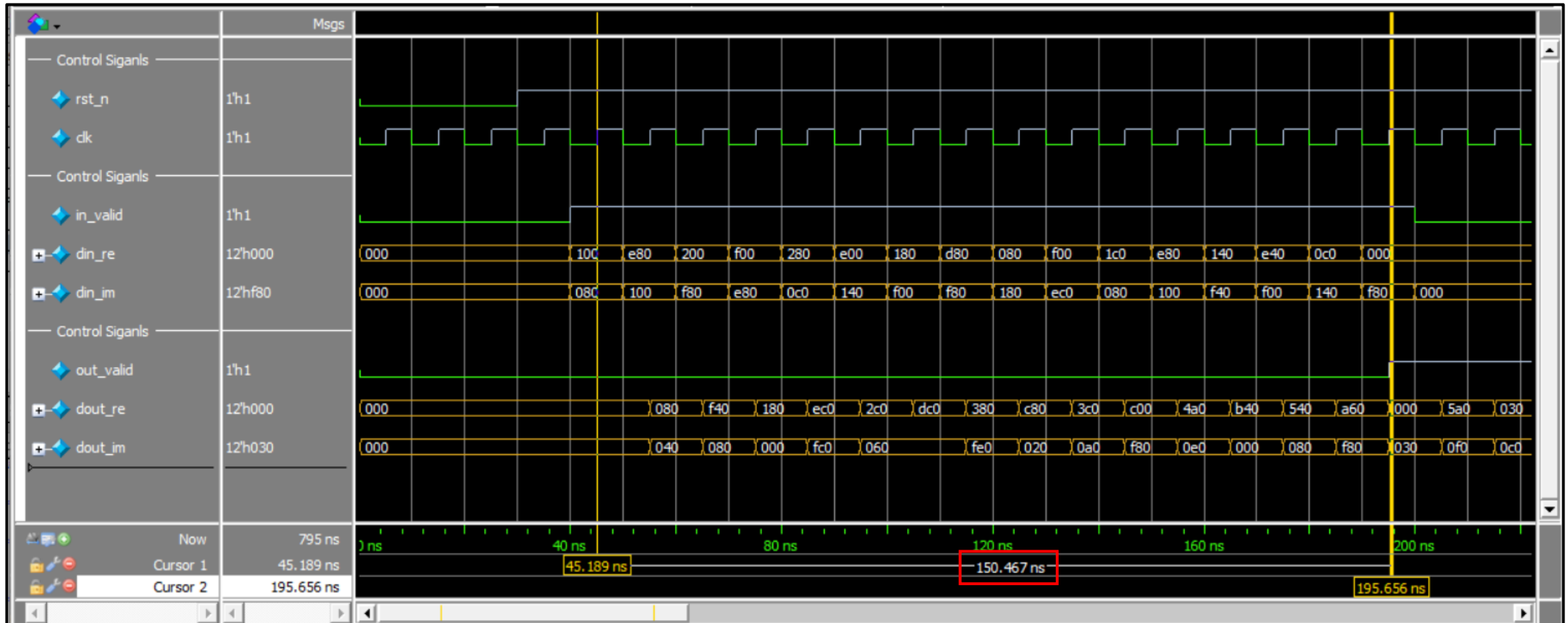
- Stage 1 : 8 cycles
- Stage 2 : 4 cycles
- Stage 3 : 2 cycles
- Stage 4 : 1 cycle
- Total Latency : 15 cycle

► Throughput:

- 1 complex sample per clock cycle (after first sample)

Top Module Latency

- Total Latency = 8 + 4 + 2 + 1 = 15 cycles



- 15 cycle each cycle take 10ns so total latency = 150ns

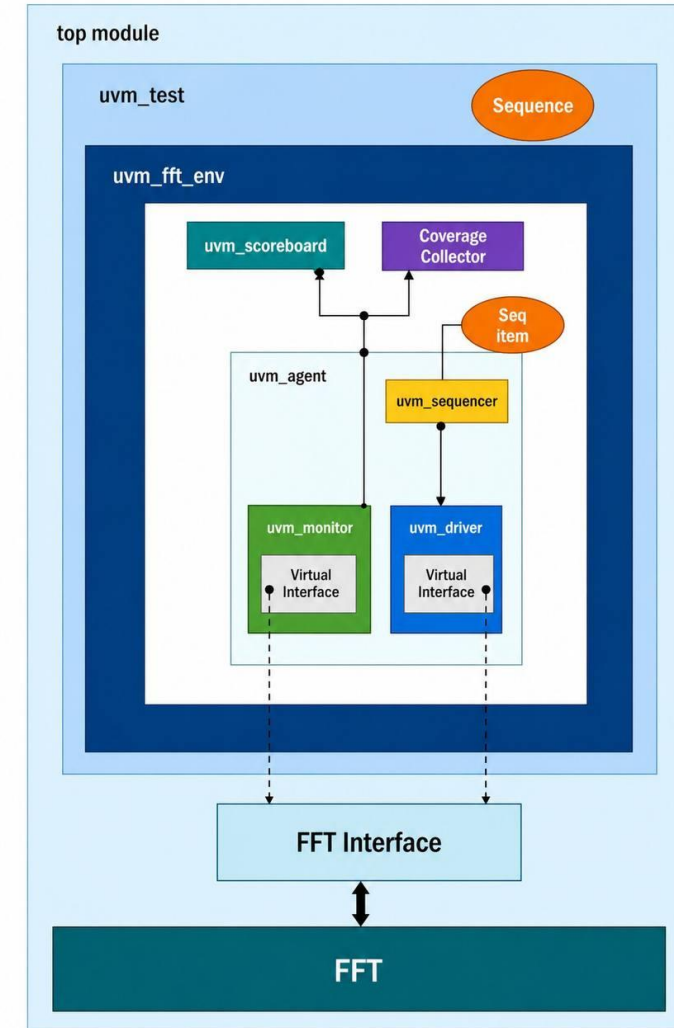
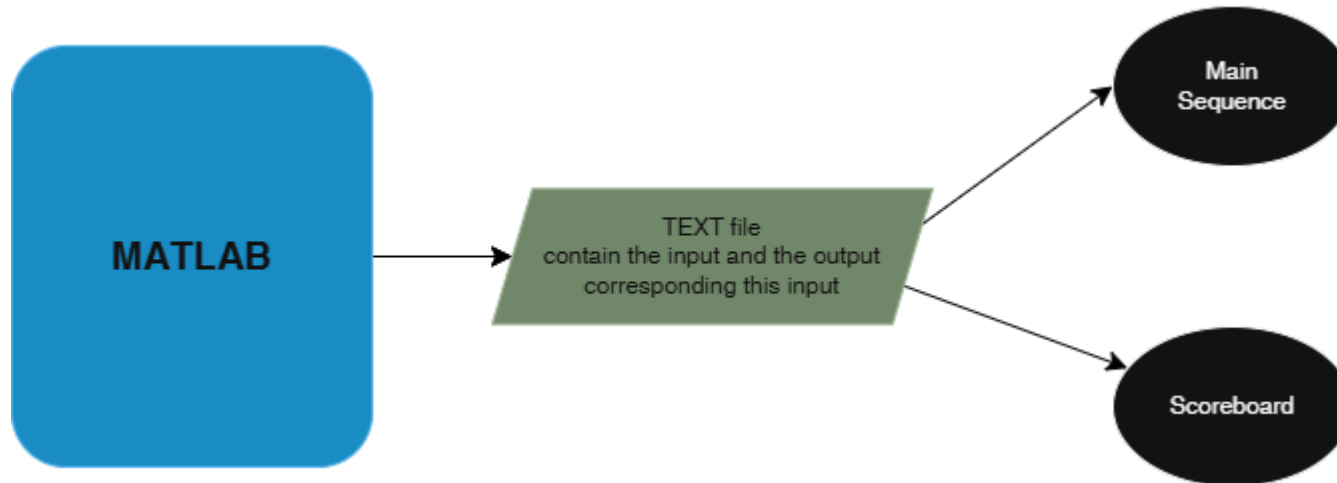
RTL Verification

Phase 3

- ▶ Comprehensive verification strategy using UVM.
- ▶ Based on end-to-end verification of complete FFT pipeline.
- ▶ Checking using either assertions or Scoreboard.
 - Scoreboard: checked the output of the data frame against expected output
 - Assertions: checked the asynchronous reset functionality and the out_valid signal also the latency (15cycle).
- ▶ Exact bit checking with MATLAB
- ▶ Generated 3 sequences to test
 - Reset seq: assert the reset signal to check reset functionality
 - Specific seq: check the behavior by entering DC signal then Impulse
 - Main seq: check that the inputs used in modeling phase in MATLAB gives the same output of the MATLAB
- ▶ Checked the **code coverage** and also made **functional coverage** to check that some sequences are happened and bins are hated or if its illegal it is not happened
- ▶ Results of the coverage represented on automated website to make reading results easier.
- ▶ Check linting the design for static RTL verification before syntesis

UVM Architecture

- ▶ Connection between classes used the TLM either port and export like between driver and sequencer or the analysis port and export like between monitor to the scoreboard and coverage collector and FIFO to wait the data to be captured
- ▶ The virtual interface passed through config database in the test class level then to lower classes level.
- ▶ No randomization used most cases are directed either manually like DC and Impulse signals or from file that has MATLAB inputs that driver force the interface with them



► UVM Report:

```
=====
                                FFT VERIFICATION REPORT
=====

Total correct samples : 16032

Data mismatches      : 0

MATLAB FFT frames    : 1000

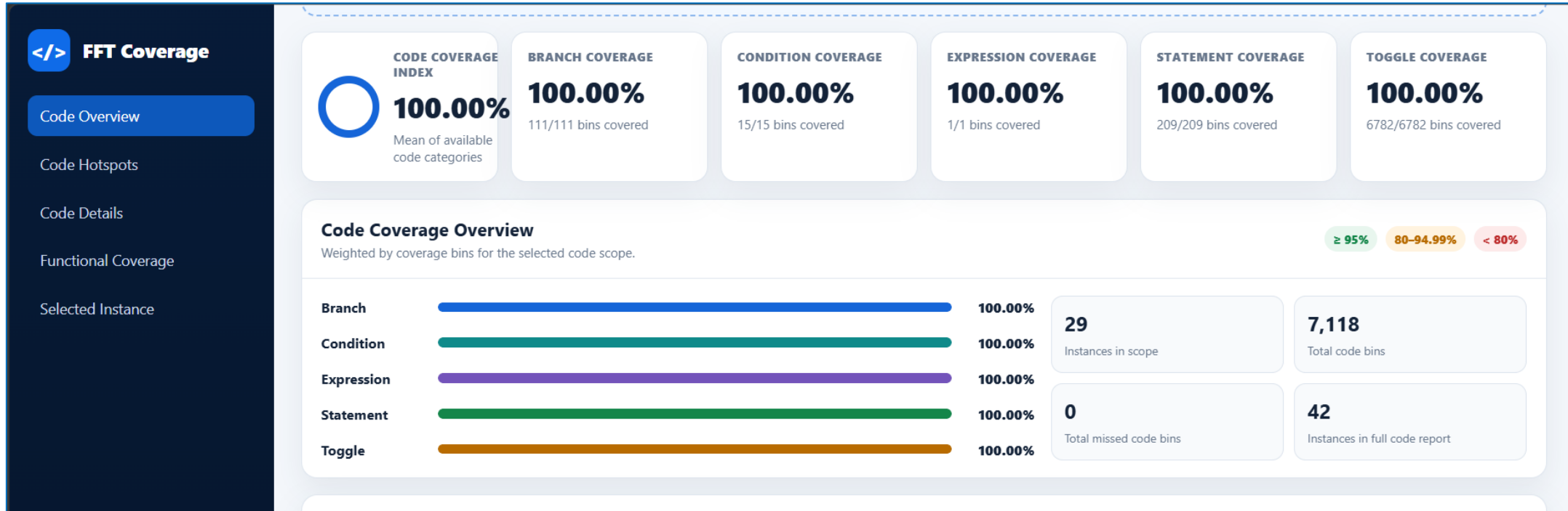
Directed tests       : DC, Impulse

=====

RESULT                : PASS

=====
```

► Code Coverage:



- Some branches like default and signal in toggle are excluded like twiddle factor signals as the are not meant to be toggled

► Functional Coverage:

TOTAL FUNCTIONAL COVERAGE

100.00%

1 covergroup type(s)

COVERPOINTS / CROSSES

7 / 2

Parsed from covergroup instances

COVERED FUNCTIONAL BINS

44/44

100.00%

IGNORE / ILLEGAL BINS

2 / 8

No illegal bins hit

/FFT_coverage_pkg::FFT_coverage::cvg_group

Covergroup instance

100.00% Covered

Reset_bins Coverpoint	100.00%	2 / 2 covered	0 missing	Covered	Bins
Valid_data Coverpoint	100.00%	2 / 2 covered	0 missing	Covered	Bins
Output_Validation Coverpoint	100.00%	2 / 2 covered	0 missing	Covered	Bins
Real_IN Coverpoint	100.00%	5 / 5 covered	0 missing	Covered	Bins
IMAG_IN Coverpoint	100.00%	3 / 3 covered	0 missing	Covered	Bins
Real_out Coverpoint	100.00%	5 / 5 covered	0 missing	Covered	Bins
IMAG_out Coverpoint	100.00%	3 / 3 covered	0 missing	Covered	Bins
In_valid_with_data_in Cross	100.00%	11 / 11 covered	0 missing	Covered	Bins
Out_valid_with_data_out Cross	100.00%	11 / 11 covered	0 missing	Covered	Bins

► Linting with spyglass:

The screenshot displays the SpyGlass IDE interface during a linting operation. The top toolbar shows the 'Run Goal' set to 'lint/lint_rtl'. The left sidebar shows the project hierarchy with 'DIF_R2SDF_top' selected. The central pane shows the Verilog code for 'top.v'.

```

1 module DIF_R2SDF_top #(
2     parameter DATA_WIDTH = 12
3 ) (
4     input wire clk,
5     input wire rst_n,
6     // FFT INPUT
7     input wire signed [DATA_WIDTH-1:0] din_re,
8     input wire signed [DATA_WIDTH-1:0] din_im,
9     input wire                in_valid,
10    //serial FFT OUTPUT
11    output wire signed [DATA_WIDTH-1:0] dout_re,
12    output wire signed [DATA_WIDTH-1:0] dout_im,
13    // Out valid

```

The right sidebar shows the 'Help Viewer' for the 'lint/lint_rtl' goal, detailing its purpose and description.

The bottom pane shows the 'Message Tree' with the following content:

Message	File	Line
Message Tree (Total: 2, Displayed: 2, Waived: 0)		
INFO (2)		
DetectTopDesignUnits (1) : Identify the top-level design units in user design.		
Module DIF_R2SDF_top is a top level design unit	rtl/top.v	1
ElabSummary (1) : Generates Elaborated design units Summary data		
Please refer file './spyglass-1/lint/lint_rtl/spyglass_reports/SpyGlass/elab_summary.rpt' for elab summary report	./spyglass-1/lint/lint_rtl/spyglass_reports/SpyGlass/elab_summary.rpt	0

The bottom status bar shows the 'Violations' tab selected, with a search bar and a dropdown menu set to 'Message Tree'.

ASIC Flow

Phase 4

- ▶ **Technology:** SkyWater130nm (sky130_fd_sc_hd -High Density library)
- ▶ **Target:** reach frequency of at least 50MHz with min area as possible
- ▶ **Constraint on time:**
 - Uncertainty for hold 0.2ns
 - Uncertainty for setup 0.1ns
 - Period 20ns
- ▶ **Area Constraints:**
 - Core utilization in placement 25% "to do high density lib higher than that make routing run fail"
 - Density 50%
- ▶ **Flow done:**
 - Synthesis -> Formality -> floorplan -> powerplan -> placement -> CTS -> Routing -> chip finishing -> signoff verification

Area report

```
1  
2 =====  
3 report_design_area  
4 =====  
5 Design area 124218 u^2 100% utilization.
```

Worst Slack report

```
1 |  
2 =====  
3 report_worst_slack -max (Setup)  
4 =====  
5 worst slack 12.32  
6  
7 =====  
8 report_worst_slack -min (Hold)  
9 =====  
10 worst slack 0.19
```

Input Files Required For Formality Check

Design Verilog Files (.v)

Netlist From synthesis (.v)

Guidance File (.svf)

Std cells libraries (.lib/.db)

Failing Point Report

```
*****
Report           : failing_points

Reference        : Ref:/WORK/DIF_R2SDF_top
Implementation   : Imp:/WORK/DIF_R2SDF_top
Version          : L-2016.03-SP1
Date             : Fri Aug 14 00:02:22 2026
*****

No failing compare points.

1
```

Unverified Points Report

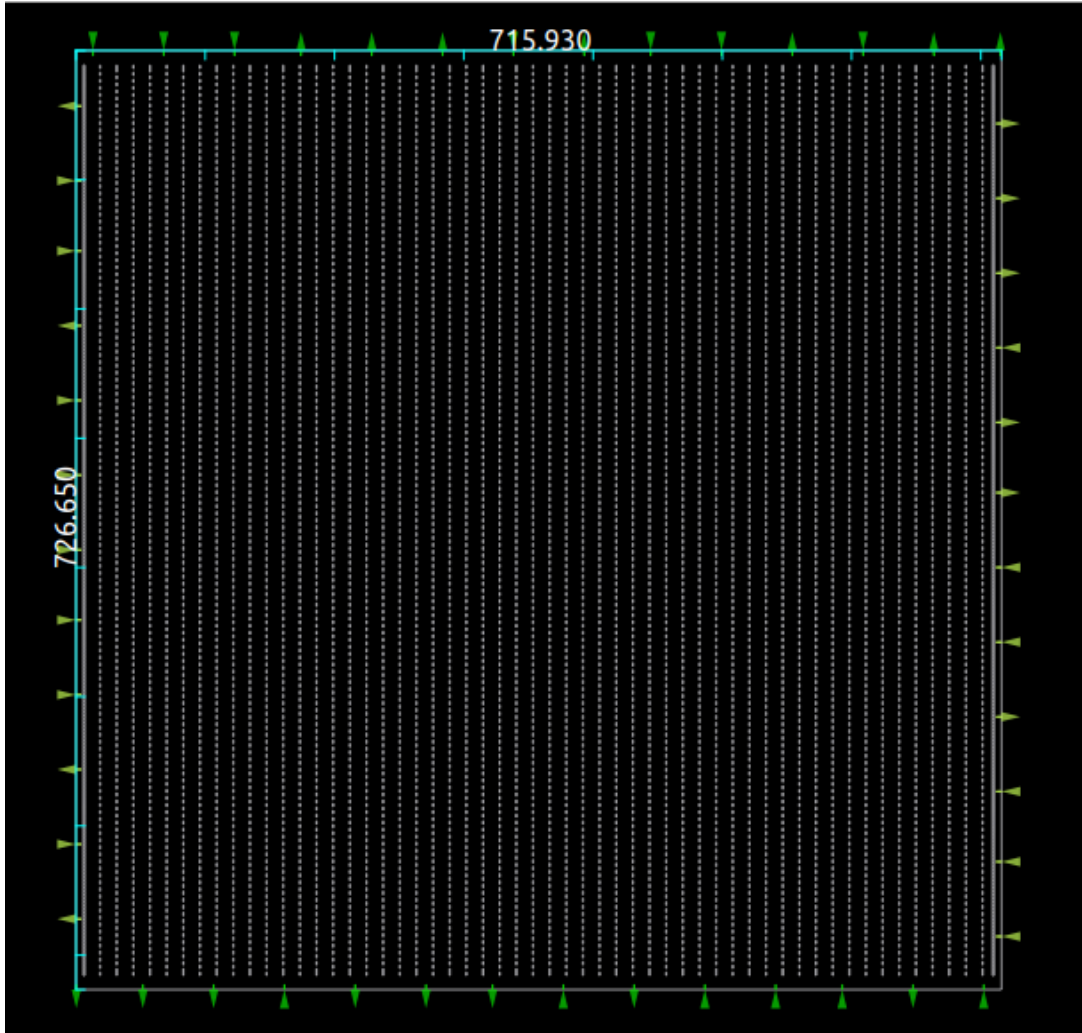
```
*****
Report           : unverified_points

Reference        : Ref:/WORK/DIF_R2SDF_top
Implementation   : Imp:/WORK/DIF_R2SDF_top
Version          : L-2016.03-SP1
Date             : Fri Aug 14 00:02:22 2026
*****

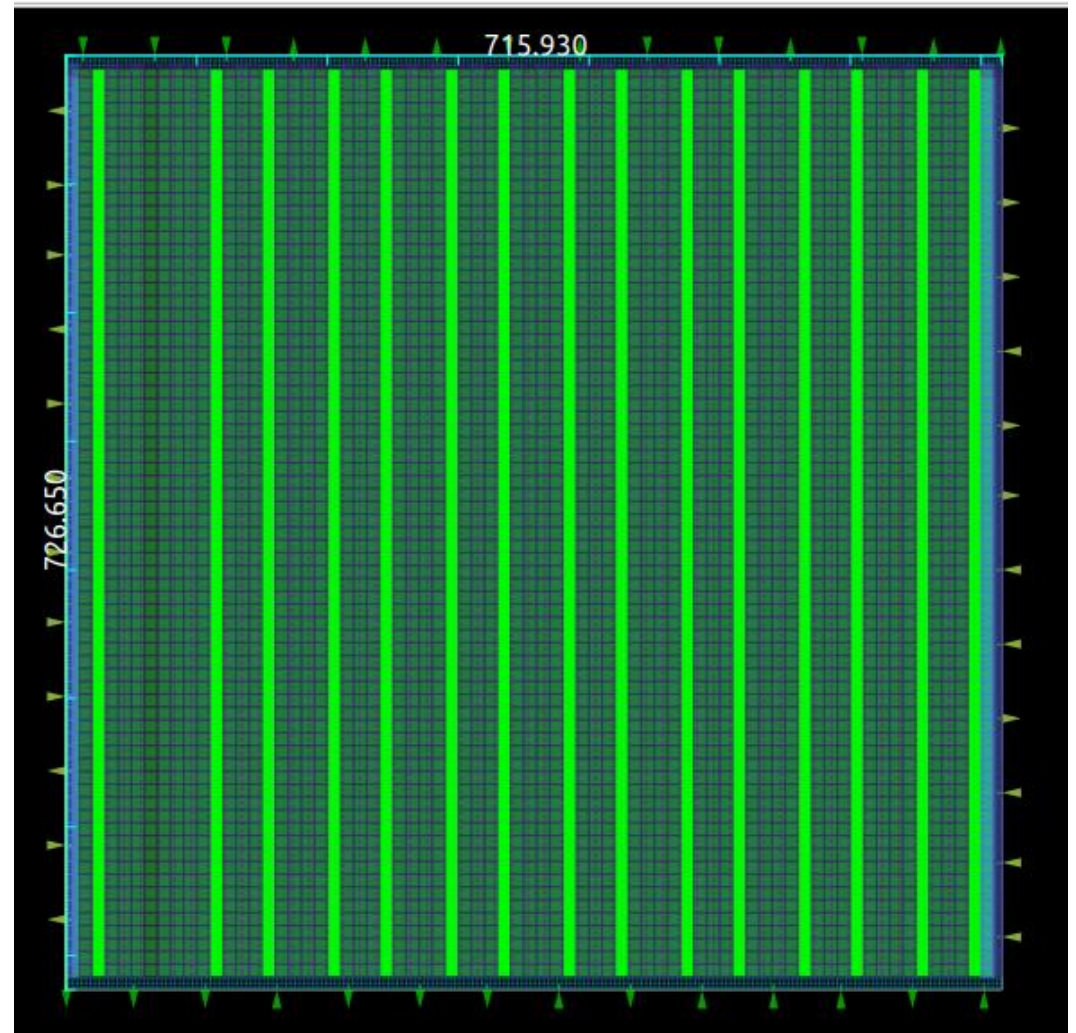
No unverified compare points.

1
```

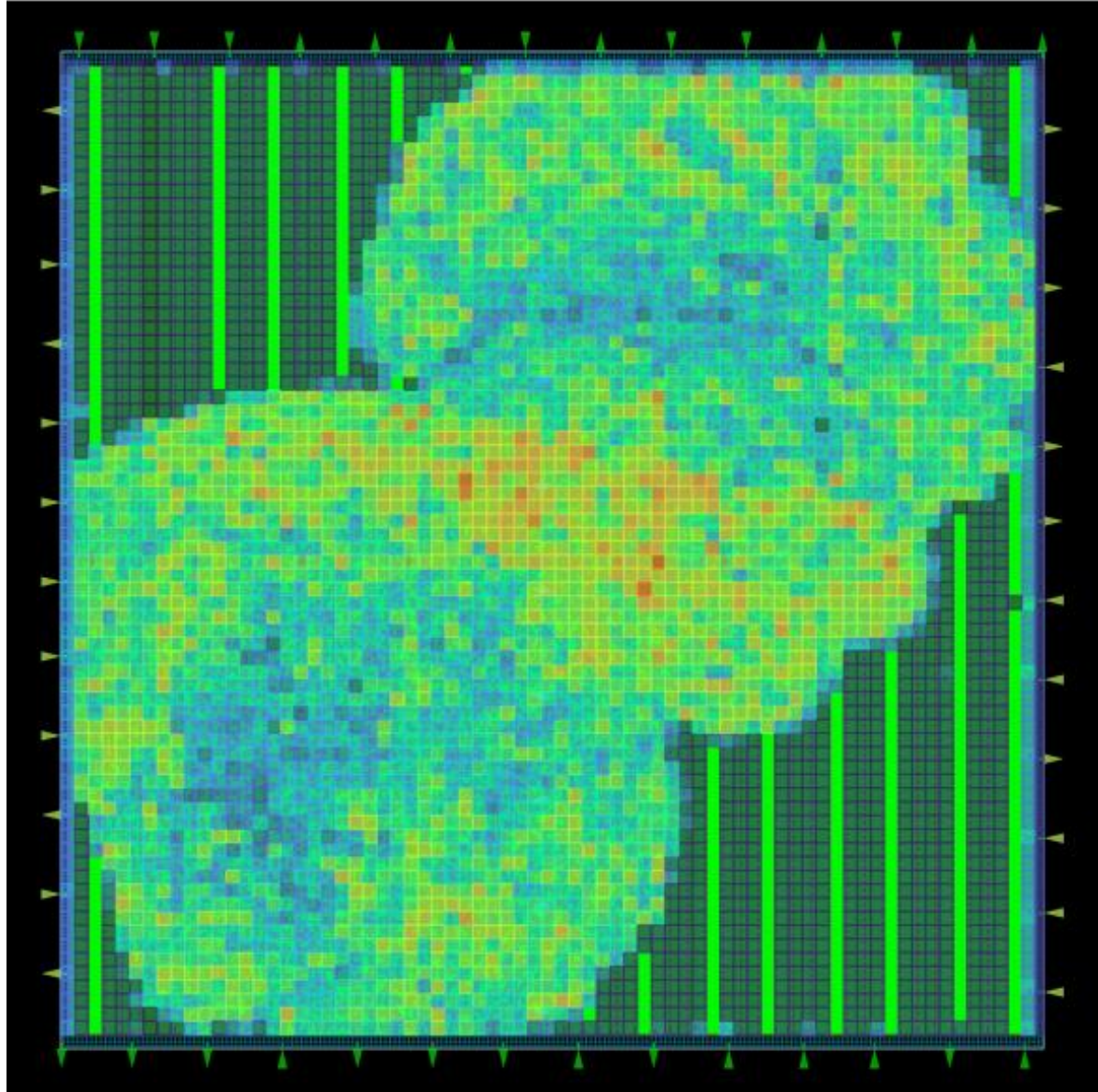
Without metal layers view

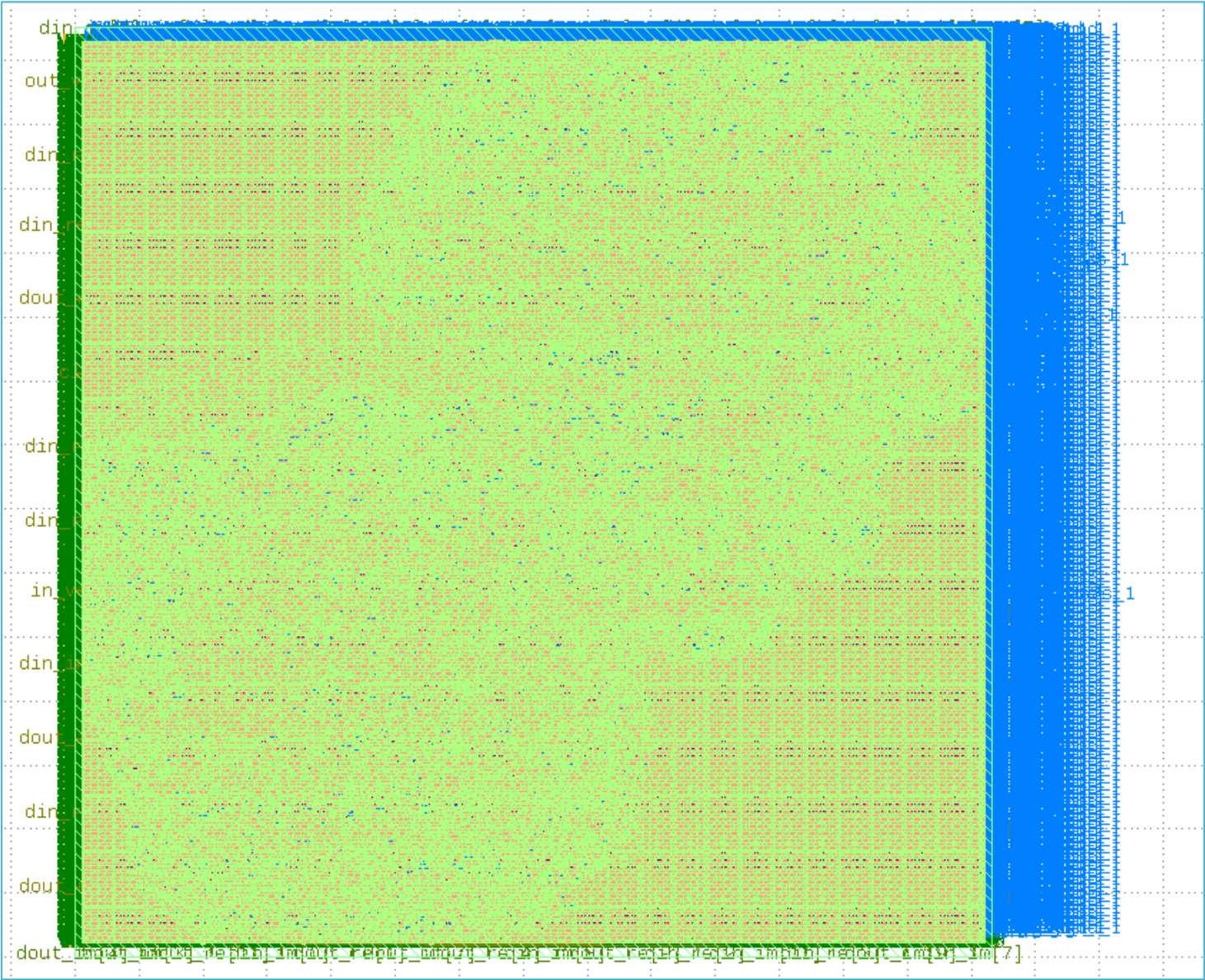


With metal layers view



Placement





Metrics	Value
Worst Negative Slack (WNS) for Setup	4.48ns
Worst Negative Slack (WNS) for Hold	0.01ns
Total Negative Slack (TNS) for both setup and hold	0ns
Max Frequency	$\frac{1}{(20 - 4.48)ns} = 64.43MHz$
Area & Utilization	483583μm ² & 97%

```
1
2 =====
3 report_design_area
4 =====
5 Design area 483583 u^2 97% utilization.
```

```
1 |
2 =====
3 report_worst_slack -max (Setup)
4 =====
5 worst slack 4.48
6
7 =====
8 report_worst_slack -min (Hold)
9 =====
10 worst slack 0.01
```

```
1 |
2 =====
3 report_wns
4 =====
5 wns 0.00
```

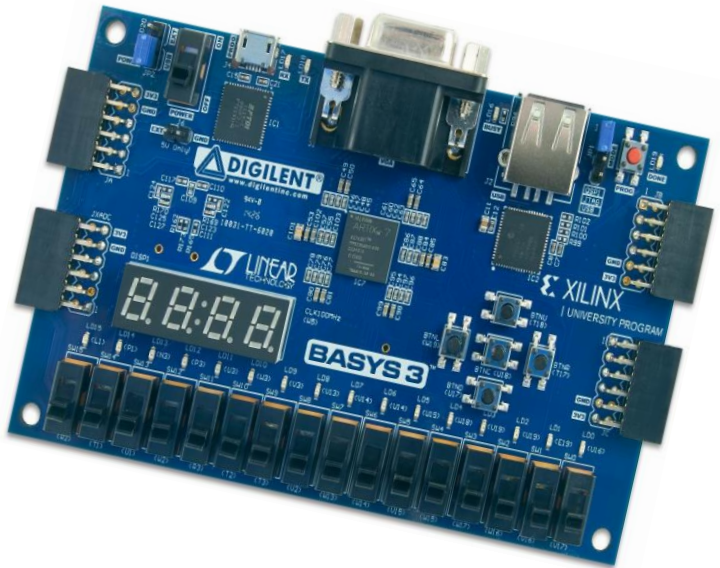
```
1
2 =====
3 report_tns
4 =====
5 tns 0.00
```

FPGA Flow

Phase 5

► Board used:

- Digilent Basys 3 board
- ARTIX-7 FPGA

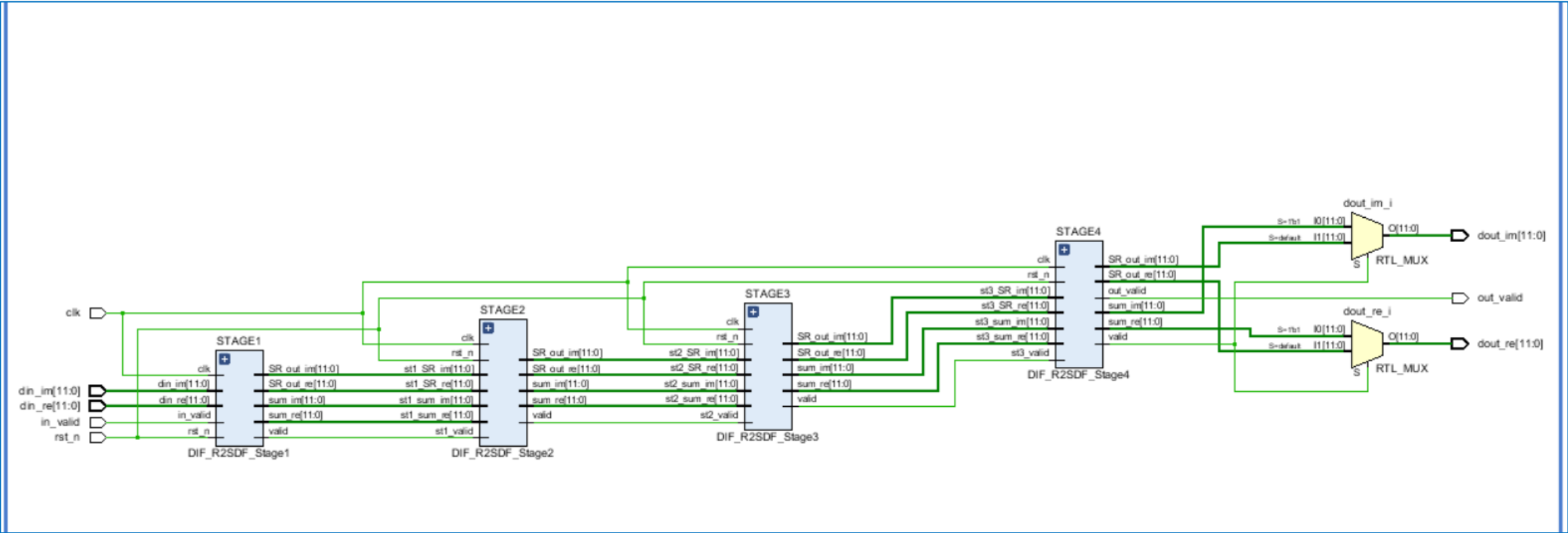


► Main constraints used:

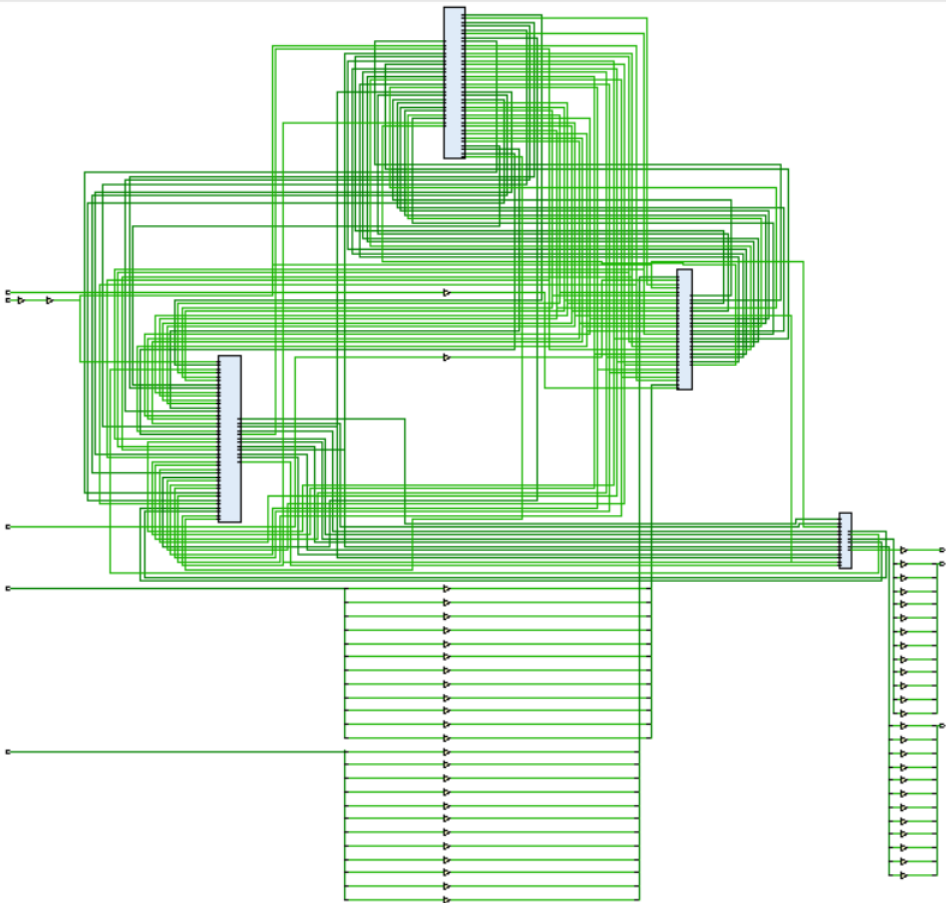
- Clk period = 20nsec

```
1  ## -----  
2  ## 50 MHz SYSTEM CLOCK  
3  ## -----  
4  
5  set_property PACKAGE_PIN W5 [get_ports clk]  
6  set_property IOSTANDARD LVCMOS33 [get_ports clk]  
7  create_clock -add -name sys_clk_pin -period 20.000 -waveform {0.000 10.000} [get_ports clk]  
8  
9  
10 ## -----
```

► Schematic:



► Schematic:



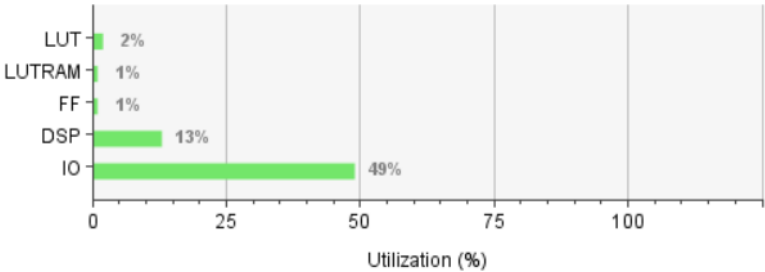
► Utilization & Timing:

Design Timing Summary

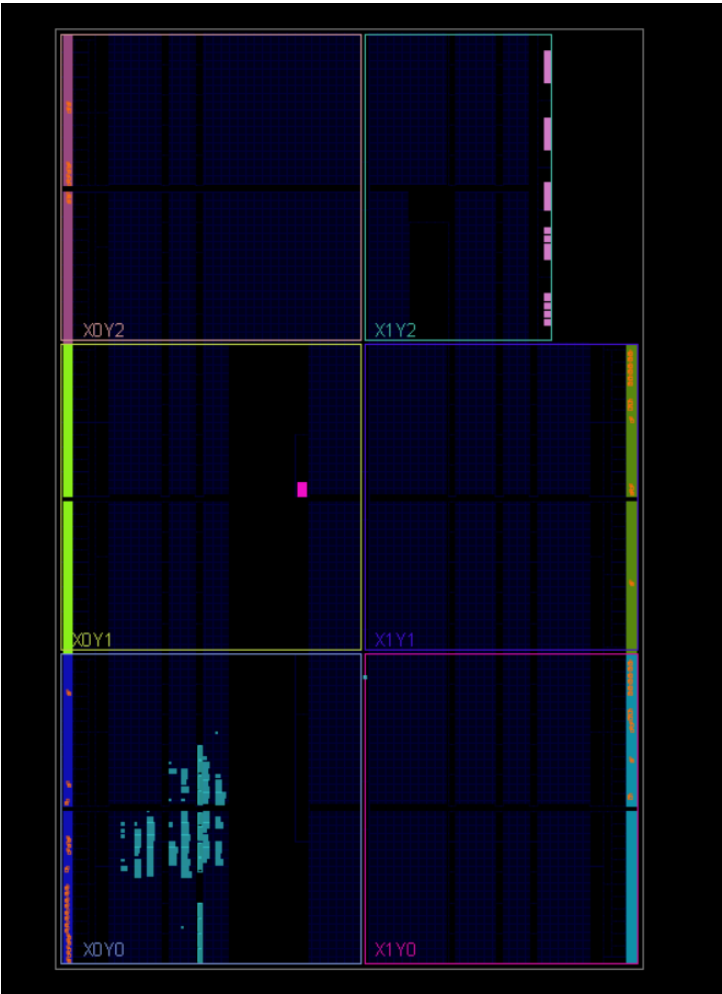
Setup	Hold	Pulse Width
Worst Negative Slack (WNS): 6.219 ns	Worst Hold Slack (WHS): 0.139 ns	Worst Pulse Width Slack (WPWS): 9.020 ns
Total Negative Slack (TNS): 0.000 ns	Total Hold Slack (THS): 0.000 ns	Total Pulse Width Negative Slack (TPWS): 0.000 ns
Number of Failing Endpoints: 0	Number of Failing Endpoints: 0	Number of Failing Endpoints: 0
Total Number of Endpoints: 269	Total Number of Endpoints: 269	Total Number of Endpoints: 276

All user specified timing constraints are met.

Resource	Utilization	Available	Utilization %
LUT	399	20800	1.92
LUTRAM	48	9600	0.50
FF	227	41600	0.55
DSP	12	90	13.33
IO	52	106	49.06



► Device:



► Utilization & Timing:

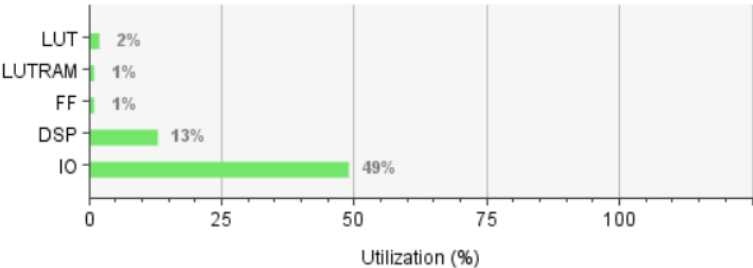
Design Timing Summary

Setup	Hold	Pulse Width
Worst Negative Slack (WNS): 3.770 ns	Worst Hold Slack (WHS): 0.126 ns	Worst Pulse Width Slack (WPWS): 9.020 ns
Total Negative Slack (TNS): 0.000 ns	Total Hold Slack (THS): 0.000 ns	Total Pulse Width Negative Slack (TPWS): 0.000 ns
Number of Failing Endpoints: 0	Number of Failing Endpoints: 0	Number of Failing Endpoints: 0
Total Number of Endpoints: 269	Total Number of Endpoints: 269	Total Number of Endpoints: 276

All user specified timing constraints are met.

Summary

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LUT	399	20800	1.92
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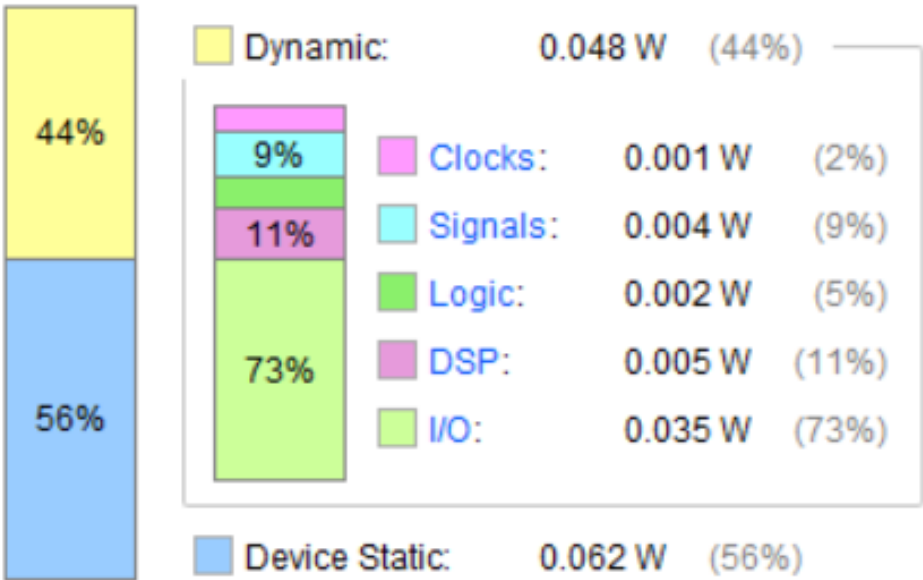


Power analysis from Implemented netlist. Activity derived from constraints files, simulation files or vectorless analysis.

Total On-Chip Power:	0.11 W
Design Power Budget:	Not Specified
Power Budget Margin:	N/A
Junction Temperature:	25.5°C
Thermal Margin:	74.5°C (14.8 W)
Effective θ_{JA} :	5.0°C/W
Power supplied to off-chip devices:	0 W
Confidence level:	Low

[Launch Power Constraint Advisor](#) to find and fix invalid switching activity

On-Chip Power



Thank You!